Question 1 **4 marks**

Define and explain the following terms

(i) Isotope

(ii) Half life

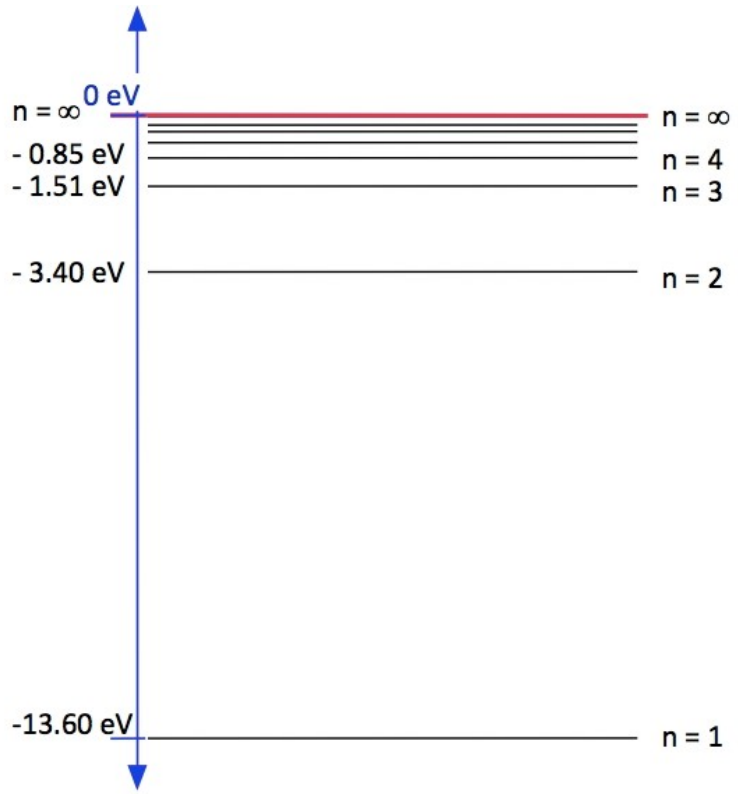
(iii) Beta irradiation

(iv) Alpha decay

[1 + 1 + 1 + 1 = 4 marks]

Question 2 **7 marks**

The diagram shows the energy of the ground state of the hydrogen atom, as well as the energies of three of its excited states.



(a) A hydrogen atom absorbs radiation so that it is excited to the third energy level above the ground state. What are all of the possible energies, in eV, of photons that could then be emitted by the hydrogen atom?

(b) Suppose in a sample of hydrogen atoms at room temperature the atoms are all in the ground state. Justify the statement that none of the radiation absorbed by any of these hydrogen atoms in the ground state is in the visible part of the electromagnetic spectrum.

[3 + 4 = 7 marks]

Question 3 **19 marks**

Americium-241, $^{241}_{95}Am$, was one of the radioactive sources used to demonstrate the properties of radiation during a PHYC10006 lecture. The source was contained in a smoke detector, and was a 3.8×10^4 Bq source of radiation.

(a) In the lecture it was shown that when a Geiger tube was placed close to the smoke detector it detected radiation – a clicking sound was observed. Then, when the Geiger tube was pulled back so that it was a centimetre away from the source, the rate of detection of radiation decreased dramatically.
What sort of nuclear radiation was emitted by the $^{241}_{95}Am$: α, β, or γ? Use the observed behaviour to justify your answer.

(b) Radon is a gas that decays via the same process as $^{241}_{95}Am$. Measurements have shown that the average concentration of radon in Australian homes is approximately 11 Bq/m^3, much lower than in most other countries. However, as it is possible for there to be variation from home to home, the Australian Radiation Protection and Nuclear Safety Agency recommends that people seek advice about how they can reduce the radon in their homes if its activity density exceeds 200 Bq/m^3.
Why is this concentration of radon a concern, while the much more active 3.8×10^4 Bq $^{241}_{95}Am$ sources in smoke detectors are not considered to be a risk to health?

The table below provides the relative atomic masses (in u) of several isotopes with similar mass numbers to $^{241}_{95}Am$.

$^{241}_{97}Bk$ 241.06023	$^{242}_{97}Bk$ 242.06198	$^{243}_{97}Bk$ 243.06301	$^{244}_{97}Bk$ 244.06518	$^{245}_{97}Bk$ 245.06636
$^{240}_{96}Cm$ 240.05553	$^{241}_{96}Cm$ 241.05765	$^{242}_{96}Cm$ 242.05884	$^{243}_{96}Cm$ 243.06139	$^{244}_{96}Cm$ 244.06275
$^{239}_{95}Am$ 239.05302	$^{240}_{95}Am$ 240.05530	$^{241}_{95}Am$ **241.05682**	$^{242}_{95}Am$ 242.05955	$^{243}_{95}Am$ 243.06138
$^{238}_{94}Pu$ 238.04956	$^{239}_{94}Pu$ 239.05216	$^{240}_{94}Pu$ 240.05381	$^{241}_{94}Pu$ 241.05685	$^{242}_{94}Pu$ 242.05874
$^{237}_{93}Np$ 237.04817	$^{238}_{93}Np$ 238.05095	$^{239}_{93}Np$ 239.05294	$^{240}_{93}Np$ 240.05616	$^{241}_{93}Np$ 241.05826

Rest masses of: $^{4}_{2}He$ atom: 4.00260 u; electron: 0.00055 u; positron: 0.00055 u.

(c) (i) Write a balanced equation for the decay process taking place.
(ii) Use this equation and the data in the table above to provide further evidence that the decay process you chose in (a) is spontaneous for $^{241}_{95}Am$.

(d) The half-life of $^{241}_{95}Am$ is 432 years. By what factor would the count rate from a smoke detector decrease after 10 years?

(e) There is 0.3×10^{-6} g of $^{241}_{95}Am$ in the smoke detector. Show that such a source would have an activity of 3.8×10^4 Bq.

[3 + 4 + 5 + 2 + 5 = 19 marks]

Question 4 **11 marks**

The table below provides the relative atomic masses (in u) of several isotopes with similar mass numbers to $^{222}_{86}Rn$. The rest masses of some common particles are also given below the table.

$^{222}_{88}Ra$ 222.01538	$^{223}_{88}Ra$ 223.01850	$^{224}_{88}Ra$ 224.02021	$^{225}_{88}Ra$ 225.02361	$^{226}_{88}Ra$ 226.02541
$^{221}_{87}Fr$ 221.01426	$^{222}_{87}Fr$ 222.01755	$^{223}_{87}Fr$ 223.01974	$^{224}_{87}Fr$ 224.02326	$^{225}_{87}Fr$ 225.02557
$^{220}_{86}Rn$ 220.01139	$^{221}_{86}Rn$ 221.01554	$^{222}_{86}Rn$ **222.01758**	$^{223}_{86}Rn$ 223.02179	$^{224}_{86}Rn$ 224.02409
$^{219}_{85}At$ 219.01243	$^{220}_{85}At$ 220.01476	$^{221}_{85}At$ 221.01559	$^{222}_{85}At$ 222.01784	$^{223}_{85}At$ 223.01914
$^{218}_{84}Po$ 218.00897	$^{219}_{84}Po$ 219.01374	$^{220}_{84}Po$ 220.01660	$^{221}_{84}Po$ 221.0	$^{222}_{84}Po$ 222.0

Rest masses of: $^{4}_{2}He$ atom: 4.002602 u; electron: 0.000549 u; positron: 0.000549 u.

$^{1}_{1}H$ atom: 1.007825 u; neutron: 1.008665 u; proton: 1.007276 u

(a) (i) Write an equation to describe the α-decay of $^{222}_{86}Rn$.

(ii) Use the data above to justify the statement that α-decay of $^{222}_{86}Rn$ is spontaneous.

(b) Calculate the average energy required to remove one nucleon from a nucleus of $^{4}_{2}He$ in an imaginary process in which the nucleus is separated into hydrogen atoms and neutrons. Express your answer in MeV.

(c) The value of the ratio $\frac{number\ of\ neutrons}{number\ of\ protons}$

for both $^{52}_{26}Fe$ (iron-52) and $^{4}_{2}He$ (helium-4, i.e. an α-particle) is 1.0 .

Helium-4 is stable, but iron-52 is radioactive. When iron-52 decays its nucleus captures an inner shell electron. Explain why the same ratio between the numbers of neutrons and protons in the nucleus results in stability for helium-4, but not for iron-52.

[(1 + 2) + 4 + 4 = 11 marks]

Question 5 **4 marks**

The isotope $^{18}_{9}F$ is used in the compound Fluoro-2-Deoxy-D-Glucose (FDG) to produce images of the body using Positron Emission Tomography (PET).

Positrons emitted from the fluorine-18 nuclei in FDG annihilate electrons and produce a pair of X-ray photons that are detected.

Use the data below to *identify* and *explain* one advantage of using a compound involving the $^{18}_{9}F$ nuclide in PET imaging, rather than the $^{11}_{6}C$ nuclide.

Isotope	Maximum positron range	Half life
$^{18}_{9}F$	2.6 millimetre	110 minute
$^{11}_{6}C$	3.8 millimetre	20 minute

[4 marks]

Question 6 **9 marks**

(a) (i) Explain why the existence of a cutoff frequency in the photoelectric effect more strongly favours a particle theory rather than a wave theory of light.

(ii) If the threshold wavelength in the photoelectric effect increases when the emitting metal is changed to a different metal, what can you say about the work function of the two metals?

(b) Two light sources are used in a photoelectric effect experiment to determine the work function of a particular metal surface. When green light from a mercury lamp ($\lambda = 546.1\,\text{nm}$) is used, a retarding potential of 1.62V reduces the photocurrent to zero.

(i) Based on this measurement, what is the work function for this metal?

(ii) What stopping potential would be observed when using the yellow light from a helium discharge tube ($\lambda = 587.5\,\text{nm}$)?

[(3 + 2) + (2 + 2) = 9 marks]